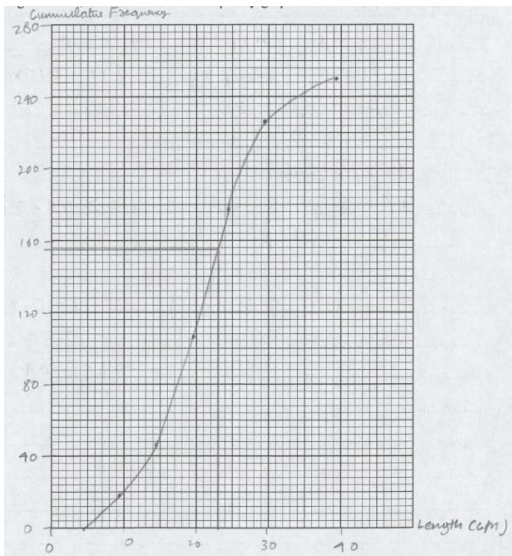




Question	Answer	Marks	Guidance										
1(a)	$\frac{1}{3} \times \frac{1}{4} \times \frac{4}{5} + \frac{1}{3} \times \frac{3}{4} \times \frac{1}{5} + \frac{2}{3} \times \frac{1}{4} \times \frac{1}{5} = \left[\frac{9}{60} = \right] \frac{3}{20}$ AG	B1	Order of coins must be consistent with question if not identified.										
		1											
1(b)	<table border="1"><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X = x)</td><td>$\frac{24}{60}, \frac{8}{20}, \frac{2}{5}$ 0.4</td><td>$\frac{26}{60}, \frac{13}{30}$ 0.433</td><td>$\frac{9}{60}, \frac{3}{20}$ 0.15</td><td>$\frac{1}{60}$ 0.0167</td></tr></table>	x	0	1	2	3	P(X = x)	$\frac{24}{60}, \frac{8}{20}, \frac{2}{5}$ 0.4	$\frac{26}{60}, \frac{13}{30}$ 0.433	$\frac{9}{60}, \frac{3}{20}$ 0.15	$\frac{1}{60}$ 0.0167	B1	Table with correct values of x and at least two correct non-zero probabilities.
	x	0	1	2	3								
	P(X = x)	$\frac{24}{60}, \frac{8}{20}, \frac{2}{5}$ 0.4	$\frac{26}{60}, \frac{13}{30}$ 0.433	$\frac{9}{60}, \frac{3}{20}$ 0.15	$\frac{1}{60}$ 0.0167								
		B1	One more correct non-zero probability linked with correct x value, need not be in table if clearly identified, accept unsimplified (total of 3 correct probabilities).										
	B1	4 correct probabilities linked with the correct outcomes, may not be in table. Decimals correct to at least 3SF. SC1 for 4 or more probabilities summing to 1 placed in a probability distribution table.											
		3											
1(c)	$[\text{Var}(X) =] \frac{(0^2 \times 24 + 1^2 \times 26 + 2^2 \times 9 + 3^2 \times 1)}{60} - \left(\frac{47}{60} \right)^2$ $= \frac{1 \times 26 + 4 \times 9 + 9 \times 1}{60} - \left(\frac{47}{60} \right)^2$	M1	Appropriate variance formula using (E(X)) ² value. FT <i>their</i> table with 3 or more probabilities (0 < p < 1) which need not sum to 1, with an expression no more evaluated than in bold . FT acceptable at the bold partially evaluated stage with <i>their</i> probabilities.										
	$= \frac{2051}{3600}, 0.570$	A1	0.5695 < Var(X) ≤ 0.570. If M0 scored, SC1 for $\frac{2051}{3600}, 0.570$ WWW Note: 0.57 without more accurate previous value penalised as 2SF.										
			2										

Question	Answer	Marks	Guidance
2(a)	$[0.55 \times 0.3 \times 0.15 \times 3! =] 0.1485, \frac{297}{2000}$	B1	Accept $\frac{148500}{1000000}$, condone 0.149.
		1	
2(b)	Method 1		
	$[1 - P(10, 11, 12) =]$ $1 - \{ {}^{12}C_{10} 0.55^{10} 0.45^2 + {}^{12}C_{11} 0.55^{11} 0.45 + 0.55^{12} \} =$ $[1 - (0.0338529 + 0.0075229 + 0.0007662) =]$	M1	One term of the form ${}^{12}C_x (p)^x (1-p)^{12-x}$, $0 < p < 1$, $x \neq 0$ or 12.
		A1	Correct unsimplified expression, no terms omitted leading to final answer. Condone omission of last bracket ‘}’ only.
	= 0.958	B1	$0.9575 < p \leq 0.958$.
	Method 2		
	$[P(0,1,2,3,4,5,6,7,8,9) =]$ $0.45^{12} + {}^{12}C_1 0.55^1 0.45^{11} + \dots + {}^{12}C_9 0.55^9 0.45^3$	M1	One term of the form ${}^{12}C_x (p)^x (1-p)^{12-x}$, $0 < p < 1$, $x \neq 0$ or 12.
		A1	Correct unsimplified expression, no more than 7 ‘middle’ terms omitted leading to final answer.
	= 0.958	B1	$0.9575 < p \leq 0.958$.
		3	

Question	Answer	Marks	Guidance
2(c)	[Mean = $140 \times 0.15 =$] 21 [Variance = $140 \times 0.15 \times 0.85 =$] 17.85	B1	21 and 17.85 (or $17\frac{17}{20}$) seen, allow unsimplified. May be in standardisation formula. ($\sigma = \sqrt{17.85}$, 4.224926... to at least 4SF implies correct variance). Withhold mark if variance clearly identified as standard deviation, condone $N(21, \sqrt{17.85})$ if standardisation formula correct or variance/standard deviation correctly stated as well.
	$P(X > 24) = P\left(Z > \frac{24.5 - 21}{\sqrt{17.85}}\right)$	M1	Substituting <i>their</i> μ and <i>their</i> σ into the $\pm$ standardisation formula (any number for 24.5), allow σ^2 or $\sqrt{\sigma}$.
		M1	Use continuity correction 23.5 or 24.5 in <i>their</i> standardisation formula. Note: If no working $\pm\left(\frac{3.5}{\sqrt{17.85}}\right)$ or $\pm\left(\frac{3.5}{4.225}\right)$ seen gains M2 BOD.
	$[P(Z > 0.8284) = 1 - \Phi(0.8284)]$ 1 – 0.7961	M1	Appropriate area Φ , from final process, must be a probability. May be implied by a sketch of the required probability area. Note: correct final answer implies this M1. Expect final answer < 0.5.
	= 0.204	A1	Final answer AWRT.
		5	

Question	Answer	Marks	Guidance														
3(a)	<table><tr><td></td><td>< 9.5</td><td>< 14.5</td><td>< 19.5</td><td>< 24.5</td><td>< 29.5</td><td>< 39.5</td></tr><tr><td>CF</td><td>18</td><td>46</td><td>106</td><td>178</td><td>226</td><td>250</td></tr></table>		< 9.5	< 14.5	< 19.5	< 24.5	< 29.5	< 39.5	CF	18	46	106	178	226	250	B1	At least 4 of 46, 106, 178, 226, 250 cumulative frequencies correct. May be implied by accurate plotting if scale suitable. May be by data table.
		< 9.5	< 14.5	< 19.5	< 24.5	< 29.5	< 39.5										
	CF	18	46	106	178	226	250										
		B1	Linearly scaled axes correctly labelled cumulative frequency (cf) (from 0 to 250), length (oe) and cm (from 5 to 39.5) – or a suitable title, with at least 3 values identified on each. Axes must be more than 50% of grid.														
		M1	At least 4 points correctly plotted at class upper end points (9.5, 14.5, 19.5, 24.5, 29.5, 39.5) on scaled axes.														
	A1	All points plotted correct, curve drawn (within tolerance) and joined to (4.5,0) and not going above 250 vertically within range. A0 if straight line segments used.															
	4																
3(b)	[250 × 0.62 = 155]	M1	Clear indication of use of graph at cf 155 is required.														
	Line drawn from 155 on cf axis to meet graph at l = 23	A1FT	Must be an increasing cf graph. Expect an answer in the range 22.5 ≤ l ≤ 23.5 from correct graph.														
		2															

Question	Answer	Marks	Guidance
3(c)	Midpoints 7, 12, 17, 22, 27, 34.5	B1	At least 5 correct midpoints seen, may be unsimplified, may be in calculation, may be by data table.
	Mean = $\frac{7 \times 18 + 12 \times 28 + 17 \times 60 + 22 \times 72 + 27 \times 48 + 34.5 \times 24}{250}$	M1	Correct unsimplified mean formula using <i>their</i> 6 midpoints (not upper bound, lower bound, upper limits, lower limits, cw, fd, f or cf and must be within class) condone 1 error. If midpoints correct accept $\frac{5190}{250}$ or $\frac{126 + 336 + 1020 + 1584 + 1296 + 828}{250}$.
	= 20.76	A1	Accept $20\frac{19}{25}$, $20\frac{38}{50}$ or $20\frac{76}{100}$, not improper fraction If M1 withheld, SC1 for 20.76 oe WWW.
		3	

Question	Answer	Marks	Guidance
4(a)	$P(WWW) = \frac{8}{16} \times \frac{7}{15} \times \frac{6}{14} \left[= \frac{336}{3360}, \frac{1}{10} \right]$ or $\frac{{}^8C_3}{{}^{16}C_3}$	M1	1 outcome seen as the product of 3 fractions with 8, 7, 6 or 5, 4, 3 or 3, 2, 1 as numerators and 16, 15, 14 or 16, 16, 16 as denominators Or 1 outcome correct in terms of combinations.
	$P(BBB) = \frac{5}{16} \times \frac{4}{15} \times \frac{3}{14} \left[= \frac{60}{3360}, \frac{1}{56} \right]$ or $\frac{{}^5C_3}{{}^{16}C_3}$	M1	Sum of 3 correct identified scenarios (may be identified by correct unsimplified numerator values).
	$P(SSS) = \frac{3}{16} \times \frac{2}{15} \times \frac{1}{14} \left[= \frac{6}{3360}, \frac{1}{560} \right]$ or $\frac{{}^3C_3}{{}^{16}C_3}$	A1	$0.1195 < p \leq 0.120$.
	$\frac{402}{3360}, \frac{67}{560}, 0.120$	3	

Question	Answer	Marks	Guidance
4(b)	3 cases to consider: WBW, WBB, WBS $P(WBW) = \frac{8}{16} \times \frac{5}{15} \times \frac{7}{14} \times 3 \left(= \frac{840}{3360}, \frac{1}{4} \right)$ $P(WBB) = \frac{8}{16} \times \frac{5}{15} \times \frac{4}{14} \times 3 \left(= \frac{480}{3360}, \frac{5}{42} \right)$ $P(WBS) = \frac{8}{16} \times \frac{5}{15} \times \frac{3}{14} \times 6 \left(= \frac{720}{3360}, \frac{3}{14} \right)$ Or $P(WBW) = \frac{{}^8C_2 \times {}^5C_1}{{}^{16}C_3} \left(= \frac{140}{560} \right)$ $P(WBB) = \frac{{}^8C_1 \times {}^5C_2}{{}^{16}C_3} \left(= \frac{80}{560} \right)$ $P(WBS) = \frac{{}^8C_1 \times {}^5C_1 \times {}^3C_1}{{}^{16}C_3} \left(= \frac{120}{560} \right)$	M1	One outcome seen as the product of 3 fractions with correct numerators and n , $(n - 1)$, $(n - 2)$ only as denominator, where $8 \leq n \leq 16$ (condone omission of $\times 3$ or $\times 6$) Or 1 outcome correct in terms of combinations with mC_3 as denominator where $8 \leq m \leq 16$. Must be a probability, no additional ‘divisions’ leading to final answer.
		A1	1 identified outcome fully correct (accept unsimplified).
		M1	Sum of 3 correctly identified scenarios (may be identified by correct unsimplified numerator values).
	$\frac{2040}{3360}, \frac{17}{28}, 0.607$	A1	If 1 or more M mark not scored, SC1 for $\frac{2040}{3360}, \frac{17}{28}, 0.607$ WWW.
	Method 2 $1 - \{P(WSS)+P(WWS)+P(WWW)+P(BSS)+P(BBS)+P(BBB)+P(SSS)\}$		
	$P(WSS) = \frac{8}{16} \times \frac{3}{15} \times \frac{2}{14} \times 3 \left(= \frac{144}{3360}, \frac{3}{70} \right)$ $P(WWS) = \frac{8}{16} \times \frac{7}{15} \times \frac{3}{14} \times 3 \left(= \frac{504}{3360}, \frac{3}{20} \right)$ $P(WWW) = \frac{8}{16} \times \frac{7}{15} \times \frac{6}{14} \left(= \frac{336}{3360}, \frac{1}{10} \right)$	M1	Two outcomes seen as the product of 3 fractions with correct numerators and n , $(n - 1)$, $(n - 2)$ only as denominator, where $8 \leq n \leq 16$ (condone omission of $\times 3$). Attempt at $1 - p$ must be present. Must be a probability, no additional ‘divisions’ leading to final answer.
		A1	2 identified outcomes fully correct (accept unsimplified).

Question	Answer	Marks	Guidance
4(b)	$P(\text{BSS}) = \frac{5}{16} \times \frac{3}{15} \times \frac{2}{14} \times 3 \left(= \frac{90}{3360}, \frac{3}{112} \right)$ $P(\text{BBS}) = \frac{5}{16} \times \frac{4}{15} \times \frac{3}{14} \times 3 \left(= \frac{180}{3360}, \frac{3}{56} \right)$ $P(\text{BBB}) = \frac{5}{16} \times \frac{4}{15} \times \frac{3}{14} \left(= \frac{60}{3360}, \frac{1}{56} \right)$ $P(\text{SSS}) = \frac{3}{16} \times \frac{2}{15} \times \frac{1}{14} \left(= \frac{6}{3360}, \frac{1}{560} \right)$	M1	1 – sum of 7 correctly identified scenarios (may be identified by correct unsimplified numerator values).
	$\frac{2040}{3360}, \frac{17}{28}$	A1	If 1 or more M mark not scored, SC1 for $\frac{2040}{3360}, \frac{17}{28}, 0.607$ WWW.
		4	

Question	Answer	Marks	Guidance
5(a)	$[P(56 < X < 75) =] P\left(\frac{56-65.8}{9.6} < Z < \frac{75-65.8}{9.6}\right)$ $[= P(-1.0208 < Z < 0.9583)]$	M1	Use of $\pm$ standardisation formula once with 65.8, 9.6 and either 56 or 75. No continuity correction, not σ^2 , not $\sqrt{\sigma}$.
	$[\Phi(0.9583) + \Phi(1.0208) - 1]$ $= 0.8309 + 0.8463 - 1$ or $0.8309 - (1 - 0.8463)$ or $0.8309 - 0.1537$ or $(0.8309 - 0.5) + (0.8463 - 0.5)$ or $0.3309 + 0.3463$	M1	Appropriate probability area Φ , from final process. Must be a probability. (expect > 0.5).
	$= 0.677$	A1	AWRT. If 1 or more M mark not awarded, SC1 for final answer 0.677 AWRT WWW.
		3	
5(b)	$\left[P\left(Z < \frac{59.1-72.4}{\sigma}\right) = 0.10\right]$	B1	1.282 or -1.282 seen cao (critical value).
	$\frac{59.1-72.4}{\sigma} = -1.282$	M1	$\pm$ standardisation formula with 59.1, 72.4, σ equating to a z-value (not 0.1, 0.9, 0.5398, 0.4602, 0.8159, 0.1841). Condone continuity correction of ± 0.05 , not σ^2 and not $\sqrt{\sigma}$. Condone $\pm \frac{13.3}{\sigma} = -1.282$.
	$\sigma = 10.4$	A1	AWRT. Signs must be consistent throughout. If M1 not awarded, SC1 $\sigma = 10.4$ WWW.
		3	

Question	Answer	Marks	Guidance
5(c)	$\left[(0.9)^4(0.1)=\right]0.0656[1]$	B1	
		1	
5(d)	Method 1		
	$[P(X < 5) =] 1 - (0.9)^4$	M1	$1 - (0.9)^d \quad d = 4, 5.$
	$= 0.344$	A1	0.3439.
	Method 2		
	$[P(X < 5) =] 0.1 + (0.1)(0.9) + (0.1)(0.9)^2 + (0.1)(0.9)^3$ or $1 - ((0.1)(0.9)^4 + (0.1)(0.9)^5 + (0.1)(0.9)^6 + (0.9)^7)$	M1	$0.1 + (0.1)(0.9) + (0.1)(0.9)^2 + (0.1)(0.9)^3 \left[+ (0.1)(0.9)^4 \right]$ or $1 - \left((0.1)(0.9)^4 + (0.1)(0.9)^5 + (0.1)(0.9)^6 + (0.9)^7 \right).$
	$= 0.344$	A1	0.3439.
		2	

Question	Answer	Marks	Guidance
6(a)	$\left[\frac{10!}{2!3!2!} = \right] 151200$	B1	CAO.
		1	
6(b)	Method 1 Total – Ys together		
	$\frac{8!}{2!2!} - \frac{7!}{2!} =$ [10080 – 2520 =]	M1	$\frac{8!}{2!a!c!} - \frac{7!}{a!b!c!}$, $a = 1, 2$ $b = 1, 2$ $c = 1, 3$.
	7560	A1	
		2	
	Method 2 3 Bs treated as a single unit $\wedge \wedge \wedge$ BBB $\wedge$ and Ys inserted		
	$\frac{6!}{2!} \times {}^7C_2$ or $\frac{6!}{2!} \times \frac{7 \times 6}{2}$ or $\frac{6!}{2!}(6+5+4+3+2+1)$ [360 × 21 =]	M1	$\frac{6!}{d!} \times {}^7C_2$ or $\frac{6!}{d!} \times {}^7P_2$ or $\frac{6!}{d!} \times \frac{{}^7P_2}{2}$ or $\frac{6!}{d!} \times \frac{7 \times 6}{e}$ or $\frac{6!}{d!}(6+5+4+3+2+1)$ $d = 1, 2, 3$ $e = 1, 2$.
	7560	A1	
		2	

Question	Answer	Marks	Guidance
6(c)	Method 1		
	$\frac{8!}{2!3!} \times 5$	*M1	$\frac{8!}{b!c!} \times d$, $b = 1,2$, $c = 1,3$ and $b \neq c$, $d \geq 1$.
		DM1	Multiply by 5.
	16 800	A1	
	Method 2		
	$\frac{{}^8P_4 \times 5!}{2!3!}$	*M1	$\frac{{}^8P_4}{b!c!} \times d$, $b = 1,2$, $c = 1,3$ and $b \neq c$, $d \geq 1$.
		DM1	Multiply by 5!.
	16 800	A1	
		3	

Question	Answer	Marks	Guidance
6(d)	Method 1		
	Y R B _ $2 \times 2 \times {}^3C_1 \times {}^3C_1 = 36$	M1	One correct identified unsimplified expression (${}^3C_1 \neq {}^3C_2$).
	Y R R B $2 \times 1 \times {}^3C_1 = 6$		
	Y R _ _ $2 \times 2 \times {}^3C_2 = 12$	B1	Correct outcome/value for 2 clearly identified scenarios, accept unsimplified WWW.
	Y R R _ $2 \times 1 \times {}^3C_1 = 6$		
		M1	Sum of 4 correct identified scenarios.
	60	A1	
	Method 2		
	Y R _ _ ${}^5C_3 \times {}^3C_1 \times {}^2C_1$	M1	5C_3 seen with YR^^ identified.
		M1	${}^5C_3 \times a, a = 2, 3, 6.$
		B1	${}^5C_3 \times {}^3C_1 \times {}^2C_1$ or ${}^5C_3 \times 3 \times 2.$
	60	A1	
		4	